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**Assessment 1 Part A - Project Proposal Submission Project
Report (Group) Part A: Project Proposal**

Course: ICT802 Capstone Project

Group Name: M52

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ICT802 Project Proposal

Working Title

Safe Service Navigation Chat Assistant for Haven House

1.2 Business and Policy Context

Haven House provides a range of community support services, including assistance for young people experiencing homelessness, mental health challenges, family crises, and involvement with the justice system. As the number of enquiries increases, staff spend significant time answering repetitive questions and directing clients to appropriate services. Haven House's mission is to help young people become safe, supported, and successful through accessible programs and timely assistance.

The proposed chatbot will support this mission by offering users immediate access to service information at any time. Rather than replacing staff, it will automate routine enquiries and guide users toward the correct services, allowing staff to dedicate more time to complex cases.

The system aligns with current trends in digital transformation within the non-profit sector by improving service accessibility while maintaining privacy, security, and ease of use.

1.3 Working Title for the Proposed Project

Safe Service Navigation Chat Assistant for Haven House (SSNCA)

1.4 Objectives

The primary objectives of the project are:

- Develop an AI-powered chatbot capable of responding to common user enquiries.

- Assist users in locating the most appropriate Haven House services.
- Reduce staff workload by automating routine questions.
- Improve accessibility by providing 24/7 assistance.
- Deliver a secure, user-friendly, and scalable web application.
- Demonstrate the practical use of Artificial Intelligence in community support services.

1.5 Project Complexity

The proposed project is considered **medium complexity**.

The system requires:

- AI-based conversational interface
- Web application development
- Database management
- Service recommendation logic
- Administrative dashboard
- User authentication
- Secure storage of service information

Although technically challenging, the project is achievable within the semester using modern development tools and frameworks.

1.6 Potential Benefits

The proposed solution provides several benefits:

- Faster response to user enquiries.
- Reduced workload for Haven House staff.
- Improved client satisfaction through immediate assistance.
- Increased accessibility outside business hours.

- Consistent delivery of accurate service information.
- Better utilisation of organisational resources.
- Opportunity for future integration with additional AI technologies.

1.7 Feasibility Statement

The project is technically, financially, and operationally feasible.

Modern AI APIs, web development frameworks, and cloud hosting platforms provide all necessary technologies to implement the proposed solution. The project scope is suitable for completion within the allocated semester timeframe.

1.8 Recommendation

Based on the identified organisational needs, expected benefits, and project feasibility, it is recommended that the **Safe Service Navigation Chat Assistant for Haven House** be approved for development.

2. Business Assessment

2.1 Situation Assessment and Problem Statement

Haven House is a community support organisation that assists individuals and families through a variety of welfare and support programs. As the organisation continues to serve a growing number of clients, staff frequently receive repetitive enquiries regarding available services, eligibility, appointment procedures, operating hours, emergency support, and referral options. Responding to these enquiries manually consumes valuable staff time and can delay assistance for clients with urgent or complex needs.

Currently, users rely on phone calls, emails, or browsing multiple webpages to obtain information. This process may be time-consuming, particularly for individuals experiencing stress or crisis, who require quick and accurate guidance. Limited office hours also reduce accessibility for users seeking information outside normal business hours.

The proposed **Safe Service Navigation Chat Assistant** addresses these challenges by providing an intelligent virtual assistant capable of answering frequently asked questions, guiding users to the appropriate services, and offering instant responses 24 hours a day. The chatbot will improve accessibility while allowing Haven House staff to focus on providing personalised support where human intervention is most valuable.

The gap between the current system and the proposed solution is summarised below.

Current Situation	Proposed Solution
Users manually search for services	AI chatbot recommends appropriate services
Staff repeatedly answer common questions	Chatbot automatically handles routine enquiries
Support available only during office hours	24/7 assistance through the chatbot
Longer response times	Immediate responses
Higher administrative workload	Reduced staff workload and improved efficiency

2.2 Options Considered

Several approaches were evaluated before selecting the proposed solution.

Option 1 – Continue with the Existing System

Maintain the current method where users contact Haven House through phone calls, emails, or the website.

Advantages

- No implementation cost.
- No additional training required.

Disadvantages

- High staff workload.
- Delayed responses.
- Limited accessibility outside office hours.
- Poor scalability.

Option 2 – Improve Website Information Only

Redesign the website and publish additional FAQs without implementing a chatbot.

Advantages

- Lower development effort.
- Improved availability of information.

Disadvantages

- Users must still search manually.
- No personalised assistance.
- Cannot provide conversational interaction.

Option 3 – AI Chat Assistant (Preferred Option)

Develop an AI-powered chatbot integrated with the Haven House website.

Advantages

- Instant responses.
- Personalised service recommendations.
- Reduced administrative workload.

- Available 24/7.
- Easy to expand with additional services.

Disadvantages

- Initial development effort.
- Requires periodic maintenance and knowledge updates.

After analysing all options, **Option 3** is recommended because it provides the greatest long-term benefit while improving service accessibility and operational efficiency.

2.3 Consultation

Consultation was undertaken with the project supervisor and project team to identify the core objectives and expected outcomes of the project. Discussions focused on improving service accessibility, reducing repetitive administrative tasks, and enhancing the user experience through digital technologies.

Feedback from these discussions indicated that an AI-powered chatbot would provide significant value by assisting users in locating appropriate services quickly while reducing the workload on Haven House staff. The stakeholders supported the concept because it aligns with modern digital service delivery practices and has the potential to improve organisational efficiency.

Future consultation will continue throughout the development lifecycle to ensure that the chatbot meets user requirements, provides accurate information, and remains aligned with Haven House's operational objectives.

3. Proposed Scope

3.1 Scope Definition

The proposed project aims to design, develop, and implement a web-based AI chatbot that assists users in finding the most appropriate Haven House services quickly and efficiently. The chatbot will provide instant responses to common enquiries and recommend suitable services based on the information provided by users.

The project focuses on improving access to information while reducing the workload associated with repetitive enquiries. The chatbot is intended to complement existing staff services rather than replace human interaction.

Project Objectives

- Develop an intelligent chatbot capable of understanding common user enquiries.
- Provide accurate information regarding Haven House services.
- Recommend suitable services based on user needs.
- Provide links and contact information for relevant services.
- Reduce response time for routine enquiries.
- Improve accessibility by offering support 24 hours a day.

Expected Outcomes

- A fully functional web-based chatbot.
- Improved navigation of Haven House services.
- Increased user satisfaction.
- Reduced administrative workload.
- Better accessibility to community support services.

Project Outputs

The project will deliver the following outputs:

- AI-powered chatbot
- Responsive web application
- Administrator dashboard
- Service management module
- Secure database
- User documentation
- Source code
- Test documentation

Quality Criteria

The project will be considered successful if:

- The chatbot provides accurate responses.
- Users can locate relevant services within a few interactions.
- The system is easy to use.
- The application performs reliably with minimal errors.
- User data is handled securely.
- The chatbot achieves satisfactory performance during testing.

Customers

The primary users of the system include:

- Haven House clients
- Young people seeking assistance
- Community members
- Haven House staff
- System administrators

3.2 Assumptions

The successful completion of the project is based on the following assumptions:

- Required software tools and development environments will be available.
- Internet connectivity will remain stable throughout development.
- AI services and APIs will remain accessible during implementation.
- Haven House service information will be available for chatbot training.
- Team members will contribute according to the project schedule.
- Users will access the chatbot through modern web browsers.
- Project requirements will remain relatively stable during development.

3.3 Constraints

Several constraints may influence the project.

Time Constraints

The project must be completed within the allocated ICT802 semester schedule.

Budget Constraints

The project will utilise free or educational development tools wherever possible to minimise development costs.

Technical Constraints

- Limited availability of premium AI services.
- API usage limitations.
- Browser compatibility requirements.
- Internet dependency for chatbot functionality.

Security Constraints

The system must ensure secure handling of user information and protect sensitive organisational data.

Resource Constraints

The project will be completed by the student project team using available hardware and software resources.

3.4 Scope of Work

The project work includes the following activities:

Included in Scope

- Requirement analysis
- System design
- Database design
- User Interface development
- Chatbot development
- AI integration
- Administrator dashboard development
- Testing
- Deployment
- Documentation
- User manual preparation

Excluded from Scope

The following activities are outside the scope of this project:

- Mobile application development
- Multilingual chatbot support
- Voice recognition functionality
- Integration with external government databases
- Online appointment booking
- Payment processing
- Live staff chat support

These features may be considered in future versions.

3.5 Implementation Strategy

The project will follow the **Agile Software Development Methodology**, allowing continuous development, testing, and feedback throughout the semester.

The implementation will be divided into the following phases:

Phase	Activities
Planning	Requirement gathering and project planning
Analysis	Business analysis and system requirements
Design	Database, UI/UX and system architecture
Development	Frontend, backend and AI chatbot implementation
Testing	Functional, integration and user acceptance testing
Deployment	Prototype deployment and demonstration
Documentation	User manual and project documentation

Estimated Resources

Resource	Details
Project Duration	Approximately 12 weeks
Team Members	4 Students

Programming Languages	HTML, CSS, JavaScript, Python
Database	MySQL
AI Technology	OpenAI API / NLP Framework
Development Tools	Visual Studio Code, GitHub
Hosting	Localhost / Cloud Deployment
Version Control	Git & GitHub

Estimated Budget

As this is an academic project, development costs are expected to be minimal. Free educational software, open-source technologies, and student licenses will be used wherever possible.

4. Project Management Outline

4.1 Governance

Effective governance is essential to ensure that the project is completed within the required scope, time, and quality standards. The project team will follow an organised management structure where each stakeholder has clearly defined roles and responsibilities.

Project Sponsor

The Project Sponsor represents Haven House and is responsible for approving the project proposal, providing business requirements, reviewing project progress, and ensuring that the developed solution aligns with organisational objectives.

Business Owner

The Business Owner represents the operational users of the system and provides feedback regarding business processes, service information, and user requirements. The Business Owner will validate that the chatbot delivers accurate and useful information to clients.

Project Manager

The Project Manager is responsible for planning, scheduling, coordinating team activities, monitoring project progress, managing risks, and ensuring timely completion of all deliverables. The Project Manager will also communicate regularly with the project supervisor and coordinate meetings among team members.

Development Team

The Development Team will be responsible for analysing requirements, designing the system architecture, developing the chatbot, implementing the database, integrating AI functionality, conducting testing, and preparing project documentation.

Project Supervisor

The Project Supervisor will provide academic guidance, review project progress during scheduled meetings, offer technical advice when required, and evaluate project deliverables according to the ICT802 assessment requirements.

Communication Plan

To ensure smooth collaboration throughout the semester:

- Weekly project meetings will be conducted.
- Progress reports will be discussed with the project supervisor.
- Tasks will be assigned according to team members' skills.
- GitHub will be used for version control and collaboration.
- Project documents will be stored in a shared cloud repository for easy access.

4.2 Key Risks and Issues

Every software project involves uncertainties that may affect successful completion. The following risks have been identified during the planning phase.

Risk	Impact	Mitigation Strategy
Changing project requirements	High	Confirm requirements early and review them regularly with the supervisor.
Delay in development	High	Follow the project schedule and monitor weekly progress.
Team member unavailability	Medium	Distribute tasks evenly and maintain clear documentation so work can be shared.
AI API limitations or service interruptions	Medium	Design the chatbot to allow alternative APIs or rule-based responses if necessary.
Technical integration issues	Medium	Perform integration testing throughout development rather than only at the end.
Database errors or data loss	High	Implement regular backups and validate data before deployment.
Security vulnerabilities	High	Apply secure authentication, input validation, and follow secure coding practices.
Insufficient testing	Medium	Conduct unit, integration, system, and user acceptance testing according to the Test Plan.
Schedule overruns	Medium	Prioritise core features first and manage project scope carefully.

Risk Management Strategy

The project team will continuously monitor identified risks throughout the project lifecycle. Weekly meetings will be used to review project progress, identify new risks, and implement corrective actions when necessary. Any significant issues affecting project scope, quality, or schedule will be discussed with the project supervisor to ensure timely resolution.

5. References (IEEE Style)

Note: Replace these with the exact references you actually use during development if required by your lecturer.

[1] I. Sommerville, *Software Engineering*, 10th ed., Pearson, 2016.

[2] R. S. Pressman and B. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed., McGraw-Hill, 2020.

[3] OpenAI, "OpenAI API Documentation," 2024. [Online]. Available: <https://platform.openai.com/docs>

[4] World Wide Web Consortium (W3C), "Web Content Accessibility Guidelines (WCAG) 2.2," 2023. [Online]. Available: <https://www.w3.org/TR/WCAG22/>

[5] Haven House, "Official Website," [Online]. Available: <https://www.havenhouse.org.au/> (or replace with the correct Haven House website specified by your project brief).